madiro

LIME EMR · TESTING

Test cases: examples

How the MSF metadata workbook becomes automated tests, with nothing new to fill in

What is a test case?

A test case is one sentence: if I do this, I should see that. It names who does it (role, site) and gives a precise, observable result. Every rule already written in the metadata workbook is a test case; we check it in the EMR automatically and report the result in the Form QA Tracker.

1. Form rules: already covered, nothing to do

Each Skip logic, Mandatory and Lower/Upper limit entry of a form tab gives one or two test cases. Examples from F17 Surgery Admission and F48 Blood Transfusion:

What the form tab says	Test case generated	Result
Skip logic: Hide question if [Referred to MSF] !== 'Yes' (on « By whom / from where »)	I answer « Yes » to « Referred to MSF » → « By whom / from where » appears. I answer « No » → it does not appear.	✅ Matches
Lower limit: 0 (on « Weight at admission (in kg) »)	I enter -1 → the value is refused.	✅ Matches
Skip logic: Hide question if [Reactions] !== 'Fever', ..., 'Other' (F48, « Management of reactions »)	I tick « Other » in Reactions → « Management of reactions » appears.	❌ Difference: the form does not show it for « Other »

Across the Release 1 workbook: about 3,900 test cases generated, 91% matching the EMR. Each difference is reviewed together: either the form or the tab is updated.

2. Another test on one question: the Comments column

For what the columns cannot express (a calculation, a plausibility check, a button), write one line per test in the existing Comments column of the form tab: Test: what I do -> what I should see.

Tab, question	Comments column
F17, Weight at admission (in kg)	Test: Enter 450 -> A warning: the weight is not plausible
F17, Order medications	Test: Click « Order medications » -> The order basket opens for this patient
F35, PHQ-9 score	Test: Answer « Nearly every day (3 points) » to the 9 questions -> PHQ-9 score = 27

3. Beyond one question or beyond forms: the User flow template tab

Patient journeys, user privileges, registration, appointments, DHIS2 and reports go in the existing User flow template tab, one step per row. Step starts with a short code; Action is what I do; Description is what I should see.

Step	Action	Description (what I should see)	User
F17-1	Save a Surgery Admission form, then open the patient chart	The admission appears in the visit with its admission date	Surgeon
ROLES-1	Register a new patient, then open their patient file	Registration works; the patient file is not accessible	Registration Clerk
ROLES-2	Open a patient file, try to order a lab test	No lab order option	MH counselor
ROLES-3	Open a patient file, go to Vitals	Vitals are visible, no « Add » button	MedCo
REG-1	Register a new patient	An MSF ID is generated: prefix IQ146-, at most 16 characters	Registration Clerk
VISIT-1	Start a visit	Visit types offered: OPD, ER, IPD, Telehealth	MH psychologist
DHIS2-1	Save an F17 form, wait for the next DHIS2 sync	The admission appears in DHIS2 with the right date	Data officer

ROLES examples come from the User roles and privileges tab: each « x » means the role can, each empty cell means it cannot, and both are tested. REG-1 comes from the Patient Identifiers tab. DHIS2-1 is illustrative.

Writing a good test

Testable	Not testable
PHQ-9 score = 27	The score is correct
No lab order option for the MH counselor role	Access rights work
I enter 450 kg → a warning appears	Weight is validated

- **Precise:** One test checks one thing, with a precise result.
- **Negative cases:** Also write what must not happen (hidden, refused, not allowed).
- **Field issues:** A problem seen in the field becomes a test, so it does not come back.
- **Data:** Never use real patient data in a test.

What would help us most

- Keep filling the form tabs as today, following the Form Guidance tab.
- Add tests in the Comments column and the User flow template tab when a rule is not covered by the columns.
- Tell us your priority forms and areas (for example roles and access, registration, DHIS2) so we review those first.